

Computer Networks.

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- ① Design & Config. star, bus, Ring, hybrid using pT.
- ② IoT based Smart home application

1.

Aim:

To Implement all topologies such as star, bus, Ring and hybrid Topology using Packet tracer.

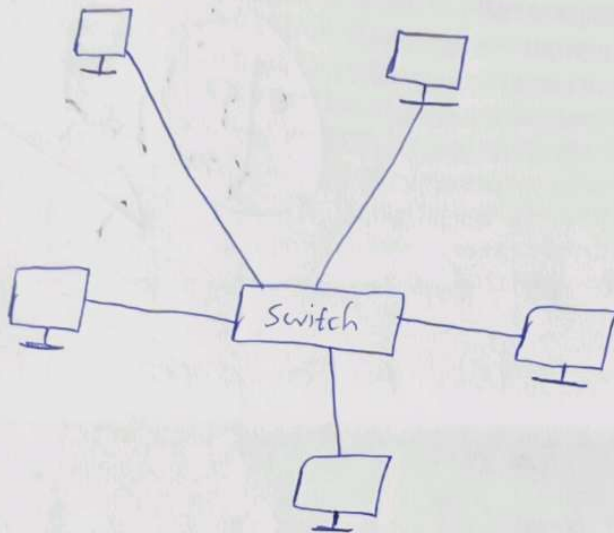
Software / Apparatus required: Packet Tracer / End devices, Hub, Connections.

Procedure:

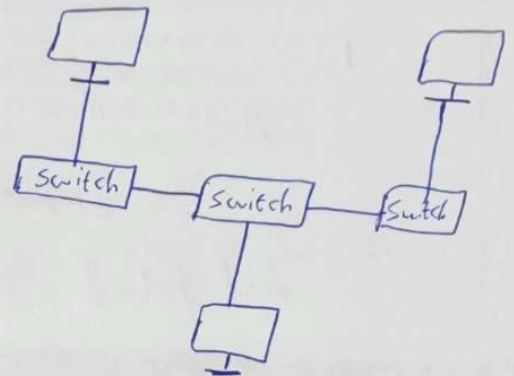
1. Start packet tracer.
2. Choose devices and connections.
3. Add Hosts and Hub.
4. ~~Connect~~ the Host and Hub with preferred wires.
5. Configure IP address and subnet mask on the Hosts.
6. Verify connectivity in Realtime Mode.
7. Verify connectivity in simulation mode.
8. Thus the Program is completed.

Topology diagrams.

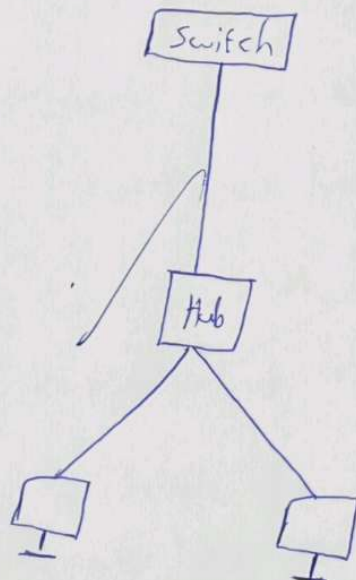
Star topology.



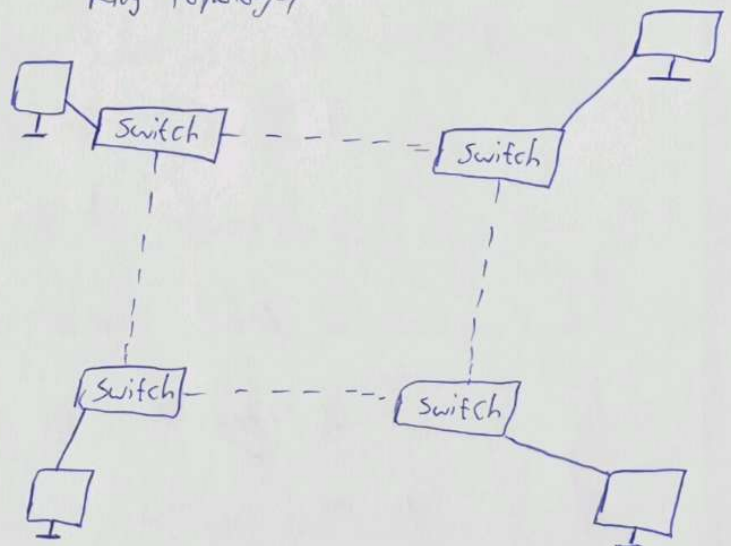
Bus Topology.



Tree Topology.



Ring Topology



Result:

Thus, the experiment is successfully created and demonstrated successfully.

2.

Aim:

To implement IoT based smart home applications in Cisco packet tracer.

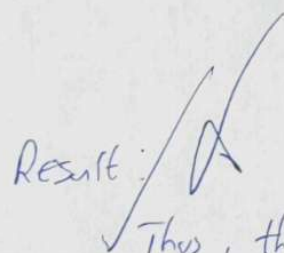
Software required:

Packet tracer.

Procedure:

1. Create a network topology in Cisco Packet Tracer that includes IoT devices such as sensors, actuators and gateways.
2. Configure the IoT devices with appropriate IP addresses, subnet masks and gateway addresses.
3. Set up a communication between the IoT devices using MQTT, CoAP, or Any other protocol of your choice.
4. Program the server to collect and send data and control the actuators.
5. Monitor and control the IoT devices via a web or mobile interface.

Result:



Thus, the experiment is successfully created and demonstrated successfully.